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ELECTROMECHANICAL EXERCISE MACHINE

Abstract

The present disclosure relates generally to a rope arrangement including a thimble; a flexible tension supporting member which is terminated on the thimble; at least one protuberance, wherein the protuberance is integral with the thimble; a channel through which the tension supporting member may travel within the channel; and a stop configured to surround the tension supporting member, such that when the stop contacts the at least one protuberance of the thimble, the stop acts as a barrier to prevent the tension supporting member from traveling beyond the stop.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/736,303 (Attorney Docket No. 36785.5) that was filed on May 4, 2022, is entitled “ELECTROMECHANICAL EXERCISE MACHINE”, and is scheduled to issue on May 27, 2025 as U.S. Pat. No. 12,311,230, which claims priority to U.S. Provisional Patent Application No. 63/184,823 that was filed on May 6, 2021 and is entitled “ELECTROMECHANICAL EXERCISE DEVICE”, which are all incorporated herein by reference for all they disclose.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] This invention relates to electromechanical exercise machines, and more particularly to a displacement-based exercise machine that allows a user to exert their maximal force output regardless of position, speed, or direction.

2. Description of the Related Art

[0003] Many modalities of muscle-building exercise are known in the art. Muscle building exercises generally are geared towards increasing peak muscle tension, increasing metabolic stress, and increasing total exercise volume. It is well-known that these factors tend to promote muscle hypertrophy. It is also well-known that the human body is capable of exerting significantly greater force in an eccentric direction (muscle lengthening) as opposed to a concentric direction (muscle contracting). The vast majority of exercise equipment is weight-based, wherein a user progressively exerts force against known static weights (commonly in the form of plates, dumbbells, weight stacks, machines, etc.). Traditional weight-based training has several significant drawbacks.

[0004] First, traditional weight-based exercise is highly inefficient because it is only able to accommodate a small fraction of the load capability of an athlete during the negative (eccentric) phase of each repetition. This is because an athlete needs to lift a weight before lowering it, so they must choose a weight they are capable of lifting. As a result, an athlete must choose a weight they are capable of lifting at their weakest position in the chosen exercise. For example, if an athlete is doing a barbell squat type exercise, they must be careful to choose a weight they can handle at the bottom of a repetition, where they have the lowest mechanical advantage. However, the same athlete is capable of supporting substantially larger loads at the top of the repetition as their mechanical advantage over the weight will be greater. Further still, an athlete will be capable of handling even more weight at the top of the repetition and in the negative (or eccentric) direction. These discrepancies between loading and athlete capability are universally present in weight-based exercise.

[0005] Second, the dynamics of weight-based exercise are the reverse of what would tend to match the force generating capability of an athlete. During the lowering (eccentric) phase of a repetition for a given exercise (when an athlete is capable of sustaining the largest loads), the apparent load an athlete feels is reduced by dynamics. The extent of this effect is dependent on how quickly the weight is lowered. Under extreme conditions, the apparent load may be reduced to zero (for instance, if the user drops the weight to the bottom). However, the apparent weight is greater during

the athlete's weakest condition in that the apparent weight is heavier during the concentric phase of each rep. The magnitude of this effect again depends on how quickly/explosively a weight is moved but may be significant.

[0006] Thirdly, weight-based exercise does not easily allow a user to adjust the resistance/weight during a set or a repetition. As described in the first disadvantage, an athlete generally must choose a fixed quantity of weight to move through exercising motions. Some weight-based or even elasticity-based exercises are available to allow weight to vary depending on the position. For example, heavy chains may be added to an exercise bar such as a barbell allowing higher loads to be suspended at the top of a repetition. Alternatively, elastic bands may be applied to a barbell similarly to provide higher tension at the top of a repetition. While this partially corrects for a change in mechanical advantage during an exercise movement, it does not allow an athlete to utilize their maximum force potential throughout the range of motion of each repetition. Both heavy chains and elastics can only provide position-dependent weight but not direction-dependent weight. Because of this, they are not capable of following an athlete's significantly higher capacity in the eccentric direction.

[0007] As a result of the above factors, there exists a need to overcome the disadvantages of weight-based or elastic-based exercise to allow an athlete to apply their maximum force on every repetition of every set. Doing so allows an athlete to stimulate higher rates of muscular hypertrophy in shorter amounts of time as compared to weight-based exercises. Many methods have been attempted to take full advantage of the human body's force generating capability, particularly including the excess force capability associated with eccentric exertion (eccentric training).

[0008] One common method of eccentric training involves transitioning between two-limbed and single-limb exercise. For instance, an athlete may perform a pull-up with two arms on the concentric phase and then lower themselves with a single arm. Or they may curl a dumbbell with two hands and then lower it with one hand. However, this method is cumbersome and may only be doable for a very small selection of exercises. Another common method is to use momentum to get a weight up and lower it slowly. Cheater reps (as they are known) are not only cumbersome, only marginally effective, and are also prone to cause injury. No weight-based or elastic-based lifting strategy allows an athlete to safely, easily, and effectively take advantage of their full lifting capacity at every point of every repetition.

[0009] Various machines have been developed which attempt to allow athletes to use their full strength potential throughout the range of motion. However, such machines still have one or more of the following disadvantages: They are prohibitively expensive. They do not provide adequate force capability. They are still force-based (or simulated weight-based) as opposed to displacement-based. They are extremely limited in that they only accommodate a single type of exercise. They present significant unresolved safety problems. Or they do not allow for mid-repetition speed and direction flexibility as they produce only isokinetic exercise.

[0010] A need exists for displacement-based exercise machines that provide greater force capacity and greater versatility while still ensuring user safety. The foregoing challenges and design considerations, as well as others, are addressed by the present invention.

SUMMARY OF THE INVENTION

[0011] Consistent with the foregoing objects and in accordance with the invention as embodied and broadly described herein, a method and apparatus are disclosed in several embodiments.

[0012] The present invention includes at least one displacement-based controller which sets the allowable position of an exercise implement. This allows a user to develop their maximum force at any or every point of every repetition. Because the present invention is displacement-based, it is capable of accepting the maximum force potential of an athlete exerting through varying mechanical advantage regardless of direction or athlete fatigue. Because the present invention is also capable of exerting a force that tracks with an athlete as they fatigue, they are also able to

produce metabolic stress much more rapidly as the machine simulates a continuously variable pyramid set in such a way that would be impossible using weight-based exercise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The foregoing and other objects and features of the present invention will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only typical embodiments of the invention and are, therefore, not to be considered limiting of its scope, the invention will be described with additional specificity and detail through the use of the accompanying drawings in which:

[0014] FIG. 1 is a perspective view of a first embodiment of the present invention, demonstrating one way that it may be used;

[0015] FIG. 2 is a perspective view of a first embodiment of the present invention;

[0016] FIG. 3 is a perspective view of a pull location as used in the present invention;

[0017] FIG. 4 is a perspective view of a first embodiment of the present invention with covers removed;

[0018] FIG. 5 is an aerial view showing the internal components of a first embodiment of the present invention;

[0019] FIG. 6A and FIG. 6B present perspective views of different positions of a safety rail mechanism as may accompany the present invention;

[0020] FIG. 7 is a perspective view of a second embodiment of the present invention, demonstrating one way that it may be used;

[0021] FIG. 8 is a perspective view of a second embodiment of the present invention with covers removed;

[0022] FIG. 9 is a perspective view of a second embodiment of the present invention;

[0023] FIG. 10 is a perspective view of a single user control which may accompany an exercise machine according to the present invention;

[0024] FIG. 11 is a perspective view of a second embodiment of the present invention, configured for transportation;

[0025] FIG. 12 is a perspective view showing the drive system of a second embodiment of the present invention;

[0026] FIG. 13 is an aerial view of a second embodiment of the present invention that shows an optional attachment for two pull-locations to be added to a single pull location;

[0027] FIG. 14 is a front view of a second embodiment of the present invention with an enhancement that allows two pull-locations to be added to a single pull location;

[0028] FIG. 15 is a flowchart describing one embodiment of a software system that may be used to control a machine according to the present invention; and

[0029] FIG. 16 is a perspective view showing an athlete performing a squat on a second embodiment of the present invention.

[0030] FIG. 17 is a perspective view of a third and preferred embodiment of the present invention.

[0031] FIG. 18 is an overhead view showing the internal arrangement of components of a third embodiment of the present invention.

[0032] FIG. 19 is a detailed view from FIG. 18 showing additional components with greater clarity from a third embodiment of the present invention.

[0033] FIG. 20 is a perspective view of a third embodiment of the present invention in an alternative configuration.

[0034] FIG. 21 is a perspective view of a kick-lift caster that may be used in conjunction with a

third embodiment of the present invention.

[0035] FIG. 22 is a perspective view of a winch and servo block that may be incorporated in a third embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0036] It will be readily understood that the components of the present invention, as generally described and illustrated in the figures herein, could be arranged and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the system (and method) of the present invention, as represented in FIGS. 1 through 22, is not intended to limit the scope of the invention, as claimed, but is merely representative of certain embodiments of apparatus and methods in accordance with the present invention.

[0037] The embodiments of systems in accordance with the present invention will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. Those of ordinary skill in the art will, of course, appreciate and that various modifications may easily be made without departing from the essential characteristics of the invention.

[0038] Referring to FIG. 1, in describing a first embodiment of an electromechanical exercise machine 100 in accordance with the present invention, it may be advantageous to first define longitudinal 101, transverse 102, and vertical 103 directions positioned to be substantially mutually orthogonal. In general, the longitudinal direction 101 will be aligned with the length of a platform portion 104 of the exercise machine 100. The transverse direction 102 will extend from side to side. By default, the vertical direction 103 will then be aligned with a direction generally up-and-down. All directions are with respect to the electromechanical exercise machine or apparatus 100. The terms exercise machine as well as exercise apparatus are used interchangeably throughout.

[0039] The structures on this exercise machine 100 typically accomplish several key functions, as will be described in detail. The primary and most fundamental function of this exercise machine 100 is to facilitate displacement-based exercise efficiently and safely so as to enable athlete strength to be developed more rapidly. This primary function is accomplished by allowing a user to perform a wide array of exercises for every major muscle group in the body.

[0040] Referring more specifically to FIG. 1 An electromechanical exercise machine 100 is generally composed of a platform portion 104 attached to a control tower section 105. A coach 106 assists a user 106 with performing exercise by manipulating controls 110 on the control tower 105. It is noted that throughout the application the term user 106 may also refer to an athlete and the term user 106 may be used interchangeably with athlete. It is further noted that throughout the application the term coach 106 is used interchangeably with the term trainer or operator. In certain instances as further described herein, a user may also be an operator of the machine 100. As depicted in FIG. 1, user 106 is performing a bench press exercise by pressing in the vertical direction on a standard Olympic barbell 109. Tension supporting members, such as ropes 153, extend to a variable length out of platform section 104 and connect to barbell 109 via two force bearing members, such as slings 132 (one on each side). Coach 106 controls the machine 100, and specifically, the length of exposed ropes 153 as it moves into and/or out of the platform 104 by manipulating the controls 110. Tension supporting members may be rods, ropes, chain, or other mechanical means and may be rigid or flexible. Force bearing members may be handles, bars, slings, or other means. It is appreciated the term ropes 153 include by way of example and not limitation, fiber ropes, cables, steel cables, cord or handles. Coach 106 controls the machine 100, and specifically, the length of exposed ropes 153 as it moves into and/or out of the platform 104 by manipulating the controls 110. A person of ordinary skill in the art will recognize that incline and decline bench press exercises (as well as a litany of other exercises) can also be performed on the present invention without departing from the scope of this disclosure. Additional exercises may be performed with force bearing members such as any standard cable attachments or barbells. Many variations are possible. It is well-known that performing a bench press exercise presents a hazard

wherein the athlete is at risk of becoming impinged between the barbell and the bench. Because the present invention is no exception to this, a set of safety rails **111** are provided, which are set at least as high as the chest height of the user **106** so as to mitigate a crushing hazard. Alternatively, this first embodiment is sized such that it will fit into a standard power rack or squat rack of ordinary design, in which case the safety rails **119** may be replaced by the safety rails of a standard rack. Safety rails **119** are connected to collars **155** which slidably engage with posts **111**. In a preferred embodiment, the post(s) **111** is/are connected to the exercise machine and oriented generally upright and generally in the vicinity of the platform **104**.

[0041] Referring specifically to FIG. 2, the electromechanical exercise machine **100** is shown without a coach or user **106**. Machine **100** includes three pull locations **116a**, **116b**, and **117** (through as few as one may also be considered), which are spaced apart in the transverse direction. [0042] As seen in FIG. 3, each pull location (**116a**, **116b** and **117**) may have a fairlead **127** situated over a pulley **128** (of ordinary design), which acts to guide a flexible tension supporting member **153** such as a rope or cord into a groove, opening or channel of the pulley **128**. The fairlead consists generally of a shaped structure with one or more contoured surfaces and is positioned such that it enables a flexible tension supporting member to be guided onto or off of a pulley. The rope or cord **153** may terminate at a thimble **130** of novel arrangement equipped with two protuberances **131**. A ball-stop **129** is situated between the thimble **130**, the fairlead **127**, and pulley **128**. This allows the rope or cord **153** to be extended from the platform portion **104** of the machine but prevents it from being retracted into the platform portion **104**. Alternative means (including traditional thimbles) may be used to terminate rope **153** without departing from the scope of the present invention. Thimble **130** is suitable for attachment to a force bearing member such as a handle, rope, bar, etc., through a carabiner (not shown) or sling **132** or the like.

[0043] Further, in FIG. 2, a platform portion **104** of the machine **100** may be equipped with a rubberized floor **118** such as is typical of gym flooring. This provides a low-impact surface that is familiar to a typical user such as an athlete. The user controls **110** on the control tower **105** preferably consist of a directional control instrument, such as a joystick **112**, an operator presence sensor **113**, an E-stop **114**, and a display device such as a screen **115**. Other arrangements of controls are possible within the scope of the present invention. For example, a set of buttons may be used as a directional control instrument in place of a joystick. In the present arrangement, the joystick **112** is provided with a spring-center bias such that activation and operation of the equipment requires continued input from the coach **106**. If the coach releases the joystick, it automatically moves back to center and all motion stops. This aids in the safety of the machine as it requires the continuous input from the coach to operate. The operator presence sensor **113** (shown as a finger sensor of ordinary design also called a touch switch) also enhances the safety of the user **106** as it further increases the requirement for the coach's presence and attention to operate the machine (i.e., both of the coach's hands are required to operate the machine). The operator presence sensor may be equipped with a time-out function so as to prevent tie-down of the presence sensor **113**, such as by permanently installing a small block into it. Thus, safety is improved because continuous operation of the controls requires continuous input from the coach, thereby requiring and maintaining the coach's presence and attention. An E-stop button **114** is also provided conveniently between the joystick **112** and the operator presence sensor **113**, which allows a coach to shut the machine down in case of emergency, for instance in the event the machine malfunctions. A touchscreen **115** may be provided with a graphical user interface for machine **100**, including workout setup, reporting real-time force and extension data, etc. Advantageously, an exercise machine according to the present invention may be configured such that the touchscreen **115** is only responsible for the user interface while the low-level functionality of the machine (especially including the safety controls) may be handled on a hardware-basis. In this way, the reliability and therefore safety of the machine can be further improved.

[0044] Referring more specifically to FIG. 4, the machine **100** is shown with the rubberized floor

118 and cover panels of control tower **105** removed to show components underneath more clearly. Floor paneling **136** is provided underneath the rubberized flooring **118**. The floor paneling **136** may be provided as standard tread grating (as shown), but other forms are also contemplated, such as sandwich paneling or the like, which may provide adequate support with light weight. The main drive input components are housed inside the control tower **105**, including a high-power motor **132**, preferably a servo motor **132** coupled to a gearbox **133**, preferably a miter gearbox **133**, which provides drive input into the platform components. The tower includes space for mounting electronics **134** and may also include a control system such as a PLC or an industrial computer **135** or other types of electronic processors. In a preferred embodiment, the control system **135** is for directing movement of an actuator **281**. A directional control mechanism for relaying a signal from the user **106** to the control system **135** may also be provided. It is understood that the signal may be an electrical, mechanical or hydraulic signal.

[0045] Referring more specifically to FIG. 5, a top-down view is shown of machine **100** with the rubberized flooring **118** and floor paneling **136** removed to show the internal components mounted in frame **137** of the platform portion **104**. FIG. 5 shows specifically how the output shaft **138** of the miter gearbox **133** is connected to the pull locations **116a**, **116b**, **117**. As shown in FIG. 5, the output shaft **138** of miter gearbox **133** is connected to a ball screw system **139**. An alternative driving means may be implemented, such as a screw, winch drive, a linear actuator, fluid power cylinder, a chain and sprocket drive, a belt drive, or any linear actuation device known in the art. The actuator, where used, allows the machine to be operated (retracting and extending the tension supporting members) in a manner other than manually. The ball screw system **139** is provided with a carriage **140**, which may travel relative to the platform portion **104** in the longitudinal direction. Carriage **140** has a first pulley **141** and a second pulley **142** connected to it. The second pulley **142** is connected to carriage **140** through a turnbuckle **143** and a force measurement device such as a load cell **144**. Alternatively, the force may be measured by, for instance, measuring the torque on the motor or other means. The turnbuckle is provided for tensioning of the system, while the load cell **144** allows the system to measure the force exerted by the athlete on the system. An energy chain **152** of typical design allows the signal from the load cell **144** to travel from the moving carriage **140** to the stationary frame **137** of the platform portion **104** and from there to the electronics mounted at **134** or computer **135** for further processing and analysis. A first rope or cord **153** and a second rope or cord **154** are provided, which functionally connect the output shaft **138** of the miter gearbox **133** to the three pull locations **116a**, **116b**, **117**. The first rope or cord **153** begins at one of the outer pull locations **116a** and passes around pulley **145** towards the carriage pulley **141**. The rope **153** next passes around pulley **146** before doubling back to carriage pulley **142**. From there rope **153** passes around pulley **147** onto sliding pulley **148**, which can slide in platform frame **137** in the longitudinal direction. From the sliding pulley, the first cord passes around pulleys **149** and **150** before terminating at the opposite outer pull location **116b**. The second cord **154**, runs from the central pull location **117** to the sliding pulley **148**. Each pulley is provided with rope retention means which prevent rope or cord from derailing from the respective pulleys in conditions where rope tension may be low. Optionally, flexible stabilizing elements **159**, **160** may be provided at the turnbuckle joints in concert with the pulley rope retention to allow the carriage **140** to be moved without an athlete exerting on any of the pull locations (**116a**, **116b**, **117**) without the rope derailing or the load string becoming unstable. This provides additional ease of use, as a user is prone to drop an attachment (for instance, when completing a set) or forget to pull on an attachment at various points in a workout. Sliding pulley **148** is capable of traveling from the position shown in FIG. 5 towards central pull location **117**, where a limit switch or a magnetic switch **151** is provided. The magnetic switch **151** may be used to indicate to the system that it has reached a limit and to discontinue the movement of the carriage **140**.

[0046] One of ordinary skill in the art will recognize the advantages afforded by the novel arrangement of the ball screw system **139**, pulleys, and ropes **153**, **154** as it allows a single actuator

to be able to control three pull locations. This provides significant cost savings on this embodiment of the present invention. The present invention is intended to be utilized by exerting on either the central pull location **117** or both of the outer pull locations **116a**, **116b**. One of ordinary skill in the art will recognize that the force seen on the load cell **144** is the same magnitude regardless of whether the athlete pulls on the center pull location **117** or both of the outer pull locations **116a**, **116b**. As an alternative embodiment, two independent drive systems may be used (one for the central pull location **117**, and one for the outer pull locations **116a**, **116b**). One of ordinary skill in the art will recognize that an independent actuator and drive system can be used for every pull location in the platform portion.

[0047] The machine is designed such that the servo motor **132** is capable of controlling the position of carriage **140** against undesired movement as force is applied at any of the three pull locations **116a**, **116b**, **117** by a user. This is due to the arrangement of the pulley system internally which functionally connects the motor to the tension supporting members at each pull location. The position of the servo motor **132** controls the exposed rope length at the pull locations (either the center pull location **117** or both of the outer pull locations **116a**, **116b**). Assuming a user is exerting against the servo, the servo's position then directly controls the displacement of the system. In practice, coach **106** may adjust the position, speed, and direction of the servo according to the athlete's needs and may even respond to verbal feedback from the athlete if desired. In this way, the entirety of every rep and every set is individually controllable, and the athlete can exert their maximum force potential through the entire range of motion. The control system which may take on the form of an on-board computer system records the displacement (from a position measurement device such as an encoder in the servo motor **132**), and force data (from the force measurement device such as a load cell **144**) collected and displays it in real-time on screen **115**. A coach may, for instance, decide to insert a pause in the middle of a given rep, allowing an athlete to settle into a position and develop their true maximum force potential. They may choose to reverse direction in the middle of a repetition to put focused work on the central portion of a repetition where the best balance between muscle tension and joint load exists. In the same set, however, the coach may still allow an athlete to train on the extreme position of a rep for a full balanced workout. All of this flexibility on-the-fly is afforded by the novel arrangement of parts as described in the present invention.

[0048] Referring particularly to FIG. 2, 6A, and 6B, the safety rails **119** are supported on vertical posts **111** with holes **123** at periodic intervals by means of collars **155** which slidably engage with vertical posts. The safety rail **119** can be positioned vertically along with the post and secured by pin **122** that goes through one of the holes **123** as well as the collar portion **155** of the vertical post **111**. The safety rail **119** includes a jog portion **120** to a rotational axis **121** which allows the support arm **119** to be toggled between a narrow and a wide position. A stopper plate **125** supports the support arm in each of these two toggle positions as it rests against a stop **156** in the upper or lower position. In the wide position (shown in FIGS. 2 and 6A) the support arm **119** is pivoted to a wide position about axis **121** which will tend to support a standard Olympic barbell **109** at a wider location that will allow an attachment means such as a sling **132** (as shown in FIG. 1) to be looped on to the barbell without interference with the rack. The support arm may also be rotated 180 degrees about axis **121** to a narrow position (as shown in FIG. 6B) that will support a standard Olympic barbell **109** at a traditional position on the outermost portion of the handle grip bar. A safety cover **124** may be provided to guard against pinch point hazards. The posts **111** may be attached to the platform portion **104** via tabs and hardware provided at **126**. A vertical stop **157** may also be provided to limit the position of the slider **155** and to prevent it from being removed. Finally, a forwardmost plate **158** is preferentially provided, which prevents a barbell from rolling off the safety rail **119**.

[0049] According to a second embodiment of the present invention, an electromechanical exercise machine/apparatus with a single pull location is disclosed. The second embodiment is generally

laid out similarly to the first embodiment. As shown in FIG. 7 the longitudinal direction **201** is similarly oriented front to back. The transverse direction **202** is oriented side-to-side, while the vertical direction **203** is oriented up and down. As shown in FIG. 7, a user **207** performs an externally loaded pushup on a standard Olympic bar **209**, which is suspended off the ground by two barbell holders **259**. The user **207** is hooked into the single pull location **216**, of the second embodiment via a force bearing member such as a shoulder sling **232** (alternatively through a standard dips belt). The length of exposed rope is controlled by a coach **206** operating controls **210** on the control tower **205**. Because there is only a single pull location the safety of the second embodiment is enhanced as compared with the first embodiment, which has three pull locations as it is more difficult to create a crushing hazard. The barbell holders **259** are at a height that makes it difficult to attach a sling or dips belt of ordinary design to the machine and be able to have enough range of motion to create a crushing hazard. As it is unlikely, though not entirely impossible, to create a crushing hazard, many other safety measures are still in place, as in the first embodiment **100**.

[0050] As shown in FIG. 8, the second embodiment machine is shown with rubberized flooring **218** removed as well as cover panels from the control tower **205**. The second embodiment machine platform portion **204** may be provided with two or more distinct segments. A first segment **261**, is situated as a central platform segment and comprises the main frame of the platform portion **204** of the second embodiment machine. Two folding wing segments (**260** and **262**) are attached to the central segment **261** through the use of hinges **263**. Alternatively, the segments may be positioned in a stowed or deployed position without folding, for instance by mechanically fastening or hooking, etc. Thus, the segments **260**, **261** or **262** may be deployed wherein the segments are positioned in a substantially planar configuration such that a user **106** may stand thereon to perform exercises or a stowed away position, wherein the segments are positioned in a manner that is more compact. Locking pins **268** may be used to lock the wing segments in the deployed (downward) position when machine **200** is in use. Sandwich panels **236** are installed in each of the three platform segments (**260**, **261**, and **262**) to provide a lightweight load bearing floor solution. The drive system is comprised of a servo motor **232** connected to a gearbox **264**, which drives a winch block **265** with a rope or cord **253** wrapped around it. Rope **253** is retained on the winch drum **265** (typically manufactured with helical rope grooves which form a channel into which the rope may travel) with the assistance of rope retention means in the form of a cylindrical keeper guard **266** which may also form a channel into which the rope may travel. Cylindrical keeper guard **266** retains rope **253** in the grooves of the winch drum **265** by providing only a small amount of clearance around the single layer of rope **253** on winch drum **265**. Rope **253** passes from winch drum **265** to a swivel pulley block **267**, where it terminates in the same manner as shown in FIG. 3. Alternative terminations that are known in the art may be used for fiber or wire rope as opposed to what is shown. Swivel pulley block **267** is equipped with rope guides **227** and a self-aligning pulley **228**. Rope guides **227** guide rope **253** into the groove or channel of pulley **228** regardless of the pull direction as the self-aligning pulley automatically aligns itself with the direction of the rope is pulled on it.

[0051] With particular reference to FIG. 9, the controls of the second embodiment may comprise a directional control instrument such as a spring-center biased joystick **212**, as well as an operator presence sensor **213**, an E-stop switch **214**, and touchscreen **215**. As shown in the second embodiment of the present invention, the touch screen **215** may be provided in the form of a hand-held tablet which can be set in a charging station **269** or removed and used from any vantage point. The versatility of this enhancement is more pronounced with the companion enhancement of a single-user mode of operation. For use in a single-user mode, the tablet may be placed in a second charging station **271**, as shown in FIG. 7. A single-user input button **270** is optionally provided on the face of the control tower **205**. The single-user input button **270** may be used to select certain options in the user interfaces as well as to enable the use of a hold-to-operate single user control to

move through the reps of individual sets. Single-user button **270** is depicted as a mechanical button, but may optionally be provided in any other electrical form, including for instance, a capacitive touch button or the like. Alternatively, a wired pendant or wireless hold-to-operate remote may be provided in various form factors to facilitate single-user operation of the machine as well. Once such embodiment (adapted as a wireless remote to clip on to a standard Olympic barbell) is shown in FIG. **10**. In FIG. **10**, a low-profile wireless remote **272** is shown, which is adapted to clip onto a standard Olympic barbell. The remote **272** is operable through the use of two buttons, **273** and **274**. Each of the buttons **273** and **274** may be configured to interact with the computer system of the single-user machine to cause the servo motor **232** to release rope **253** for extension or to draw it back in to control tower **205**. For safety purposes, remote **272** may be configured as a hold-to-operate control in that the machine will not move during a set unless either control button **273** or **274** is continuously pressed. The hold-to operate control may be implemented with redundant buttons in either direction (more particularly in the downward direction) with continuous self-checking to further increase the reliability and safety of the machine. Clip **272** may be placed on a barbell at a location where it may be easily operated for instance, by a thumb. Other embodiments of a single user control are also contemplated as they may be suitable for other exercises. For example, a foot switch may be designed for use in exercises that require both hands. Such a foot switch may be configured in a hold-to-operate manner such that a leftward input corresponds to a downward movement of the machine and a rightward input corresponds to upward movement of the machine. Further, the buttons on the hold-to-operate single user control may be configured to be feathered and be only pressed one at a time, either mechanically or electrically. Alternatively, a low-profile wireless remote **272** (or wired pendant) may be provided as a single user control with a single force sensitive resistor (FSR) or capacitive/inductive touch pad input. Various commands can be transmitted by the athlete to the machine via gestures input to the FSR including a single click, a double click, and various holding pressures, all of which can be translated into individual commands. For instance, a single click (tap) could be used to indicate to the machine to set a limit for the range of motion while the exercise is in the setup phase. A double click (tap) can be used to indicate that the athlete is ready to begin the set. A 3 second sustained hold, during the setup phase, can be used to indicate that the limits need to be reset. Once the athlete has begun the set, a 0-2 oz holding force may be used to indicate the control is not being used and the machine may be configured in an E-stop condition. A 2-4 oz force may be used to indicate operator presence. A 4 oz-15 lb force may be used to indicate that the set can proceed at full output capacity. And anything greater and **151b** input force may be used to create an E-stop state. A Person of Ordinary Skill in the Art will recognize that inputs and gestures can be mapped in many ways without departing from the scope of the present disclosure and the preceding scheme is only exemplary.

[0052] With particular reference to FIG. **11**, the utility of the segmented platform (if provided) is illustrated. As shown in FIG. **11**, the foldable wing segments **260** and **262** may be folded upward using handles **277** and attached to the control tower **205** through the use of hasps **278**. The mobility of an embodiment of the present invention may be enhanced by adding a transport system with at least one wheel that is removably mounted to the platform. For example, three jack-stands **276** of ordinary design may be mounted to lifting brackets **275**, which are bolted to portions of the center platform panel **261**. Jack-stands **276** may be used to lift the single-pulley machine, at which point it may be rolled around for easy mobility. Foldable wing segments **260** and **262** of the platform portion **204** allow the entire machine to become narrow enough to fit through a **36** in wide opening. Jack-stands **276** may be used in a similar fashion to adding a mobility enhancement to the first embodiment as well. Alternatively, as will be shown subsequently, the jack-stands can be replaced with kick-lift casters. As a further alternative, wheels may be permanently attached to the machine apparatus. A person of ordinary skill in the art will recognize that many possibilities exist.

[0053] With particular reference to FIG. **12**, a detailed view of the drive system of the second embodiment is shown. The drive system comprises the servo motor **232**, the gearbox **264**, the

winch drum **265**, and the drive block frame **284**. The drive block frame **284** is pivotally attached to the control tower frame **279** at a pivoting pin **280**. A force measurement device such as a load cell **244**, is provided, which is connected on one side to the drive block frame **284** and on the other side to the control tower frame **279**. This arrangement creates a relationship between a force exerted on winch line **253** and a force seen on load cell **244**, thereby allowing the user input force to be measured. A locking plate **283** may be provided with locking pawl **282** operated by an actuator **281** such as a rotary actuator or rotary solenoid. When the rotary actuator **281** is activated, it moves the locking pawl **282** towards the locking plate **283** thereby allowing a force to be supported on extension line **253** without the continuous activation of servo motor **232**. This enhancement provides several advantages in that it allows the single-pulley machine to continue to support loads in the event the servo motor **232** overheats. It also allows the single pulley machine to support greater loads mechanically than the servo motor **232** may be otherwise capable of generating, thereby allowing for high forces with a lower cost servo motor **232**. This locking system may also be implemented on a machine such as the first embodiment machine **100**.

[0054] Additional enhancements and embodiments are also contemplated. For instance, an electromechanical exercise machine, according to the present invention, may be constructed with one or more pull locations. Embodiments with a single pull location and three pull locations have been described in detail. However, a two pull-location embodiment may also be constructed in various ways. For example, the center pull location of the three-pulley machine may be eliminated, thereby providing a two pull-location machine. Additional pull locations may be added through the use of external accessories or external structures. For example, an external pulley system may be added to the single-pulley machine **200** such that it provides the option and versatility of being used with two pull locations. Many possibilities exist; for instance, FIGS. **13** and **14** show one such possibility. Rope or cord **253** of the second embodiment may be extended out and around an externally mounted pulley **302**, which may be mounted below the surface of the platform **204**. The external pulley **302** may be equipped with a proximity sensor on pin **303**, which disables the use of a single user mode for increased safety when two-pull locations are available. After running around an external pulley **302**, rope **253** may terminate at a coupling **301** which is attached to two smaller ropes, **304a** and **304b**. Rope **304a** may pass around pulley **305** and **309** before exiting at an external pull location **310**. Similarly, rope **304b** may pass around pulleys **306** and **307** before exiting at a pull location **308**. One of ordinary skill in the art will recognize that the arrangement of FIG. **13** allows a single-pulley machine to be adapted for use as a two-pulley machine. As shown in FIG. **14** optional tower accessories **311** may be installed on platform section **204**, which provide variable vertical positioning of pull locations **314**, **316**. If towers such as these are incorporated with a three pull location design (as will be shown subsequently) this will result in a five pull location machine. Optionally, a pullup bar of ordinary design (not shown) may be installed between tower accessories **311**. As shown in FIG. **14**, ropes **304a** and **304b** of FIG. **13** exit platform section **204** in an upward direction. Rope **304a** may pass around pulley **313** and terminate at an attachment **316** (shown as a handle) of any desired type. Pulley **313** is slidably connected to a tower **311** via a safety rail **315a**. As a matter of further enhancement of safety, rope **304a** may pass through the safety rail **315a**, thereby ensuring that the safety rail must be used in this configuration. In a similar manner, rope **304b** may pass through safety rail **315b**, around pulley **312**, and terminate at attachment **314**. One of ordinary skill in the art will recognize the advantages of the arrangement of FIG. **14** in that it provides extensive versatility in the types of exercises that can be performed. Optionally, a more complex arrangement of pulleys may be implemented with the tower accessories shown in FIG. **14** such that a constant length of rope is required regardless of the vertical positioning of the safety rails **315a** and **315b**, such as is well-known in the art.

[0055] One of ordinary skill in the art will recognize the utility and versatility of the exercise methods that can be accomplished using the apparatus disclosed above. The apparatus fills a long-felt need for a displacement-based exercise machine that is low-cost and provides variability and

customization on a per-repetition basis. A displacement-based exercise machine allows for maximum force potential to be developed by a user at every point of every repetition, especially on the eccentric phase of a repetition where a user's force potential is highest. Further, displacement-based exercise allows significantly greater flexibility on each repetition as compared with simple isokinetic exercise. Specifically, displacement-based exercise with a hold-to-operate control by either the user or the coach allows for pauses (where isometric portions may be inserted) as well as reversals mid rep (which may be centered around peak kinematic potential). These sorts of elements may be efficiently and easily inserted into any rep at any time in a set that also allows full utilization and exercise through the full range of motion.

[0056] A software system may accompany the apparatus previously disclosed to further enhance the versatility and safety of the machine. FIG. 15 shows a flowchart that demonstrates one possible way software can be used to accompany the apparatus disclosed above. As shown in FIG. 15, any of the embodiments disclosed, or other variations which do not depart from the present invention may be configured to be operable through the use of a software-based graphical user interface, mediated, for instance, through a touchscreen or the like. After the present machine is turned on (“Start Machine” **401**), an additional layer of safety may be implemented through the use of a user login **402**. In this way, the use of the machine may be limited to those who have been trained in proper and safe use. As shown, the interface may be configured such that it will check the login credentials against a known set of credentials (as in **403**) and only proceed to load the main page **404** once a user provides valid credentials. The credentials may be a password only but may also have a username and password. A username and password requirement or the use of different passwords for different users can provide further functionality in that alternative operational modes can be associated with different credentials. For example, a normal use mode may be provided in contrast to an admin mode.

[0057] Once at the Main Page **404**, a user may select from one or more modes of operation. As shown in FIG. 15, a “quick start,” **405** “baseline,” **406**, and a “training” **407** modes are provided. A “quick start” mode, if provided, may be configured to permit a user to perform exercises quickly but without a specific training benchmark of any kind or without recording of data or being associated with any specific exercise. In contrast, a “baseline” training mode **406** may be provided, which may read a “I rep max” (an athlete's maximum capacity for a single repetition) for a specific exercise performed by a specific athlete. A data storage system may be provided to recall or export collected baseline data for further use and analysis. As yet another mode of operation, a “training” mode **407**, if provided, may be configured to allow a user to train relative to their baseline or “I rep max” for a specific exercise. For example, a user may wish to train at perhaps 70% of their “I rep max.” To accomplish this, a training mode may be provided which allows a user to select the previously recorded “I rep max” data for a specific exercise performed by a specific athlete and scale the recorded data to 70% (optionally +/-some tolerance band to create a target channel). Data collected during a “training” mode may also be stored and recalled or exported for further use and analysis. For instance, historical data may be used to determine how much progress an athlete has made in a certain amount of time. It may also be used to determine if progress has plateaued for any reason and thereby assist with a diagnostics process. In any operational mode, force and displacement data may be displayed visually to an athlete (for instance, on a plot) as it is being collected in real-time. When recalled from previous training sessions (for instance, during training), a previous exercise plot may be loaded and displayed (or a scaled version of it), and, newly collected data may be plotted over the top of it in real-time such that an athlete may visually adjust their effort relative to their previously recorded “I rep max.” This may help an athlete, for instance, to pace their effort so as to produce greater exercise volume. Additional modes of operation are possible. For instance, other modes of operation are contemplated, which may simulate weight-based exercise but where the weight may change depending on position, direction, speed. In this way, greater flexibility may be achieved with the exercise and operation of the present invention.

Functionally there is a great deal of similarity between the three operational modes highlighted. Supposing that a user selects the “quick start” mode of operation, the system may be configured to present a warning page **408**, which brings some of the hazards of exercise and displacement-based exercise specifically to the user's attention. As is a typical practice, an exercise machine is provided with warning decals to alert the user to hazards and encourage them to use the machine safely. However, it is well-known that a user may not always read or fully comprehend a warning decal prior to use. Further, a user may not be particularly safety conscious. As an additional layer of safety associated with the present invention, a warning decal or simple instruction may be presented to the user before enabling any movement thereof. The user may be required to press a button in a user interface to “accept” **409** that they understand the dangers of exercise and specifically with displacement-based exercise. This makes it more likely that a user will see and understand the warning information.

[0058] After accepting a warning **409**, the system may be configured to load a setup page **410**, which allows a user to input desired parameters for the specific exercise they wish to perform. For example, a user may be performing a workout with a coach, in which case they would press select the “trainer mode” **428**, which signals to the system to enable the tower controls and keep the single user controls disabled once the workout starts. Alternatively, they may select the “solo” or “single-user” mode **411**, in which case the control tower controls will be disabled, and the single user control will be enabled during the set. From here, a user may input various set parameters **412**, **429**. For instance, the set may be defined by a number of reps, a time to travel up/down, and a top/bottom position to stop at. Set parameters may be different between a single user mode **411** and a trainer mode **428**. For instance, if the present invention is configured to auto-move during a set, additional set parameters may be provided for single-user modes, such as a start delay (in seconds), an isometric pause at the top/bottom (in seconds) and other parameters as necessary. While a time up/down may be selected, the motor can only go, so fast so the possibility exists that a user could select a time up/down that the machine can't keep up with. As an alternative, a travel speed up/down may be provided as an input that is limited to the capability of the motor.

[0059] As an additional layer of safety, the present invention may be configured to require a user to set a top and a bottom position prior to performing a set. As a user is unlikely to know to the inch where the top and bottom positions should be for a given exercise, the machine may be configured to measure it for a user. This may be done by enabling the controls of the machine to be used with a low force limitation to allow a user to move the machine to the top and press a “top” button in the user interface, and then move the machine to the bottom and press a “bottom” button in the user interface. This provides an additional layer of safety because this information may be used to restrict the movement of the machine to the range of motion a user is capable of safely moving through. Thereby it prevents hyperextension and attempted machine travel to dangerous locations. For instance, if a user is performing a bench press, this setup process will require them to set the bottom above their chest. By setting the bottom in this way, the machine movement may be limited, thereby never attempting to move to a position below the user's chest which may be dangerous. In this way, movement of the machine may be stopped regardless of continued user input.

Additionally, movement limit thresholds as described serve to provide a very consistent range of motion from repetition to repetition. Alternatively, the bottom and top positions may be measured automatically by moving the machine in direct response to tensile input from the user, simulating, for instance, the performance of a tape measure tool of ordinary design. The machine may seek to draw in under minimal tension, which can easily be supported by a user. If tensile input from a user exceeds a small threshold, the machine may allow the ropes to extend. In this way, the position of a user may be tracked by the machine in real-time. Once positioned in one of the extremes, a user or coach may press an input such as the “top” or “bottom” button in the user interface to signal to the machine that the current position should be treated as the top or bottom of a repetition. During the top/bottom measurement process, the machine output force may be limited to a safe value. The

machine may be configured to disable starting a set until all parameters have been set.

[0060] Once all the input parameters are set, a user may press a “start” button **413, 430** to begin an exercise cycle. Once an exercise cycle has been started, user controls (either the control tower or the single-user controls) may be enabled such that by manipulating them, a user (either a trainer or an athlete) can cause the machine to move depending on the machine mode. The machine may be configured to load a set screen **414, 431**, which may display current set information as desired. For example, a set screen **414, 431** may be configured to display a real-time plot of position and force as a graphical element. Further, a set screen can be configured to display running statistics such as set time, total work/volume, peak force output, and other parameters as desired. As shown in FIG. **14**, the set may be run as a repeating loop type structure in which the machine continuously cycles through scanning inputs **417, 434**, responding to inputs, and updating an output display **415, 432**. In this manner, within the capability of the machine, all display and machine response will be continuously up-to-date. Such a continuous loop structure is well known in programming, and a person of ordinary skill in the art would readily understand that a loop-type structure may take different forms and sequences while accomplishing the same objective, namely keeping inputs and outputs up-to-date. As an additional alternative, an event-based programmatic structure may be used in which rather than continuously cycling through inputs/outputs, event listeners may trigger the system to respond to individual inputs as necessary. This approach has the advantage of reducing computing power demand. Many types of programmatic control may be implemented. The control-loop type structure disclosed in FIG. **14** is only exemplary in nature but may, for instance, begin with updating output plots and stats **415, 432** with the most current information. If the user interface is configured to plot force and displacement, for instance, on a common X-Y plot, this step will cause the current data point to be added to the plot. Alternatively, current force information may be displayed, for instance, as a bar or column chart, simple text numerical display, or more artistically as a dial indicator such as a scale. Many variations are possible. However, regardless of which display method is implemented, this step will keep this output updated. After completing this step, the machine may be configured to check if a set is completed **416, 433**. While this step will never return a YES output on the first iteration of the loop, as the machine is continuously performing this step, it will keep the information display up-to-date. The set may be configured to be complete (in which case this would return a YES output) in multiple ways, two of which are disclosed in FIG. **14**. First, a set may be completed when the machine has moved through the set range of motion the set number of times (reps). At this point, the user will have completed the amount of work that they entered into the machine before beginning their set. Alternatively, a cancel/abort option **425, 441** may be provided, which would allow a user to skip reps and automatically end a set if necessary. For instance, depending on user output, they may reach the desired level of muscle fatigue sooner than anticipated or may have realized that they want to change some of the input parameters on a new set. A back button may be provided at every point in the sequence, thereby allowing easier navigation of the user interface as well. If a set is determined to be complete, the machine may be configured to exit the set control loop and load a “stats” page **426, 442**. The stats page may display various parameters of interest for the set, including peak force, total volume, etc., for an athlete and/or coach to review. Once done reviewing the final set data, the user may choose to end the set cycle and go back to the main screen **427, 443**.

[0061] If the set is not determined to be complete yet, the next step in the control loop may be to scan the inputs **417, 434**, consisting of the user controls, the load cell, and the encoder. By continuously performing this step, the machine is able to continuously keep up-to-date with the inputs and respond accordingly. Consequently, the next steps in the control-loop consist of making decisions based on these inputs. For instance, with the trainer mode, the machine may be configured to check the state of the operator presence sensor **418** first. This provides an additional layer of safety as unless the operator is present, no other machine movement will occur. As the single user controls may be configured to rely on redundant self-checking hold-to-operate buttons,

no operator presence sensor may need to be checked when using this mode. Next, the position of the joystick or single user control buttons may be responded to. If the joystick is pressed forward **419** or the up button is pressed on the single-user control **435** (depending on which control set is enabled), the machine may be configured to check to see if it is at or past the upper limit **420, 436**, in which case it will not move any further despite the user input to the contrary. Instead, it will cycle back to the top of the control loop. If, however, the machine is not at or past its upper limit, this would be determined to be a valid input from a user, and the machine will move in the direction indicated by the user **421, 437**. In the same way, if the joystick is pulled back **422** or the down button is pressed **438** the machine may test to see if it is at or below the lower limit **423, 439**. If it is, then no movement will occur, and the loop will be restarted. If it is not, then the machine may be configured to move in the indicated direction **424, 440**.

[0062] It will be apparent to a person of ordinary skill in the art that the present system may be used to safely implement displacement-based exercise. As shown in FIG. **15**, a hold-to-operate control style may be implemented to provide maximum safety and exercise versatility. Other modalities of control may be implemented. For instance, instead of a hold-to-operate control, the machine may be configured to move between extremes automatically, without requiring continuous user input. In this case, as long as the user does not change position between setting the top and bottom and completing the set, they can remain safe for the duration of a set and have the added convenience of having the machine move automatically through the set. Various pre-programmed movement patterns (including pauses and/or reversals) may be implemented.

[0063] Many other software-based or hardware-based features may be implemented to provide further enhancements to the functionality of an electromechanical exercise machine, according to the present invention. For instance, the electronic display and control system may be configured to be able to broadcast or otherwise propagate its display on a secondary screen of arbitrary size. For instance, accompanying the embodiment of the three-pulley machine, a secondary screen may be connected and configured to display set data while a user is performing a set. This direct visual feedback can help the user to know how hard to push and thereby help muscular development take place faster. A similar setup may be used for the embodiment of the single pulley machine. As an additional utility, a secondary screen may be prominently displayed such that individuals other than the coach or user may see the progress of an exercise. In this way, such a screen may be utilized for marketing purposes.

[0064] As yet another software-based enhancement, if a baseline or training mode is provided as previously disclosed, a machine according to the present invention may be configured to store data locally as well as remotely such that a user may be able to use any number of machines in the same gym or abroad and load their data. This data may optionally be tied to a specific profile in a manner as is well-known. An external interface through an internet connection may also be provided where a user may see additional analysis of their data and download or export their data for further use.

[0065] Yet another enhancement contemplated by the present disclosure is the capability through programmatic control to compensate for the change in length of a rope. This has the advantage of providing means whereby greater variety in rope materials may be accommodated. It is well known that steel cable is commonly used in exercise equipment. In typical weight-based exercise equipment, the loading is only a few hundred pounds at most (typically) and so the steel cable required to support that load may be small. However, when higher loads are used, such as may be used in an embodiment of the present invention with a significantly higher force capability the size of the steel cable and therefore associated pulleys may become prohibitively large. Larger steel cables are also known to have reduced flexibility and therefore tend to require large pulleys to maintain appropriate minimum bend radii. This may pose a significant design challenge as the components to place and guard become larger. However, fiber ropes are well-known to provide significantly improved flexibility over steel cable as well as very high strength-to-weight ratios. Further, fiber ropes are not prone to create significant recoil hazards in the case of a cable failure as

steel cables are. Fiber ropes may therefore be used to support very large loads safely while allowing much smaller pulleys to be used. However, many fiber ropes are liable to change length under tension due to a tightening of the rope lay and/or elastic stretch. This can cause problems when the top and bottom limits of a set are determined at low tension, and the exercise is performed at high tension. In order to correct for this, fiber ropes that are not prone to lengthening may be used, or software-based compensation strategies may be employed. By modeling the length/tension curve for the rope, the output of the machine can be adjusted to provide the simulation of a completely inextensible rope. This software-based compensation, if provided, needs to be consistent with the structural layout of the machine it is used with. For instance, the compensation routine for the first embodiment **100** according to the present invention will necessarily have to be different than the compensation routine for the second embodiment **200** according to the present invention.

[0066] Several further enhancements in terms of instrumentation and control are also contemplated. Referring specifically to FIG. **16**, a user **207** is depicted as performing a squat on a single-pulley machine **200**. This squat exercise is facilitated through the use of a belt **285** (which may be a dips belt) attached to a rope **253** which can be extended from or retracted into machine **200**. User **207** can exert their entire mechanical capacity against rope **253** through every repetition of every set of squats. A hand-held remote **291** is provided with one or more buttons **292** and **293** which a user may use to control machine **200** without the need for a coach. As a first operational enhancement, user **207** may wear a flexion sensor **287**, shown for examples as attached by straps **286** and **288**. Flexion sensor **287** may be configured for example as a strain-gauge type sensor which may be used to measure knee joint articulation. Data collected from flexion sensor **287** may be collected (wirelessly or through a hard data line) by machine **200** and plotted along side other data for use in calculating performance metrics. Flexion sensor **287** may be used to provide additional safety in that a warning may be issued by the machine **200** if joint articulation reaches an unsafe position. Flexion sensor **287** may also be used to improve consistency on every repetition as well as to gauge muscle action and balance. Depending on the posture and movement of user **207**, a squat may be used to focus on the quadriceps muscle or the glutes. Flexion sensor **207** may be adapted to measure the balance between cooperating muscle groups. Optionally, multiple flexion sensors **207** may be used simultaneously thereby providing additional insight. Flexion sensors such as **270** may optionally be adapted for use on a hand of user **207** whereby user gestures may be interpreted and used to send commands to machine **200**. As a further enhancement, a biometrics measuring device **290** attached to a wristband **289** may be provided which may be configured to measure athletic performance metrics such as heartrate, skin conductivity, joint articulation, etc. Data generated by such a measurement device may also be collected by machine **200** and use for informational display and calculations of various performance metrics. Biometrics measuring device **290** may also be configured to send commands to machine **200** in place of a remote **291**. As yet a further enhancement, machine **200** may be equipped with a microphone, and may be configured to accept and interpret voice commands from user **207**, thereby negating the need for a remote **291**. As a further enhancement consistent with the present invention, machine **200** may be provided with a touchscreen **215** which may be placed in a charging station **269** or a front-facing charging station **271**. The software system of machine **200** may be configured to display passive or interactive training assistance content on touchscreen **215** which may be visible to user **207** when it is placed in front-facing charging station **271** (or any location where the touchscreen **215** faces user **207**). This content may include, for instance, videos of personal trainers performing exercises and/or workouts, motivational feedback in the form of audio (music, speech) and visual elements (points, awards, banners, and the like). In addition to these features, a mobile app may optionally be provided which allows users to see and track their personal progress.

[0067] With reference to FIGS. **17** and **18**, a third and preferred embodiment is shown in a perspective view. In particular, a three-pulley machine **500** is shown in the context of a longitudinal

direction **501**, a transverse direction **502**, and a vertical direction **503**. The machine **500** comprises a platform portion **504** and a control tower portion **505** and may be used by an user **507** working alone or in conjunction with a coach **506**. The machine **500** is generally supported on a set of feet **524**. As shown in FIG. **17**, user **507** is performing a bench press exercise using a barbell **509** being pulled on by tension supporting members, in this case, ropes **554b** and **554c**. The machine **500** is provided with at least three pull locations comprising two outer pull locations **516a** and **516b** as well as a center pull location **517**. The machine **500** may comprise an athletic surface **518** such as a rubberized floor which may include a pattern or design **521** for aesthetic purposes. As shown in FIG. **17**, machine **500** may be provided with posts **511a** and **511b**. Posts **511** are supported by rear braces **525** as well as pull-up bar portion **527**. Optionally, the pull-up bar portion **527** may be provided with wide grip handles **528** and hammer grip handles **529** or any other common bar pattern used for pull ups. User **507** may use pull-up bar **527** by itself or in conjunction with machine **500**. Collars portions **526** slidably engage with posts **511** and may be selectively locked at varying heights through the use of locking pins **522** which engage with corresponding holes **523** in the posts **511**. Collars **526** may mount support cups **519**, which have a channel or passage for the tension supporting member to pass through. In use, a user **507** may perform many exercises using machine **500** including a bench-press exercise as shown in FIG. **17**. When performing a bench-press exercise, support cups **519** must be set at a proper height above the chest of user **507** and securely locked in place using pins **522**. Rope portions **554b** and **554c** pass through respective support cups **519** towards barbell **509**. Thus, in use, machine **500** may only pull barbell **509** downward toward user **507** until it engages with support cups **519** which are provided with adequate structural capacity to support the full pulling capacity of machine **500**. A person of ordinary skill in the art will recognize that this arrangement provides a safe manner in which a user **507** may perform high capacity bench-press exercises without risk of serious injury. Machine **500** may be used without posts **511a** and **511b** for various exercises as well including squats. During a squats exercise for instance, user **507** may stand over center pull location **517** in a manner similar to what is depicted in FIG. **16** and may optionally grasp handles **520** for added lateral support during a squats exercise.

[0068] With particular reference to FIG. **18**, the internal arrangement of the machine **500** is shown. A person of ordinary skill in the art will recognize that many different types of actuators may be provided to run machine **500**, however, machine **500** is shown utilizing a winch system, similar to what is shown in FIG. **12**. This winch system includes a winch drum **565** (under retaining covers), a servo motor **532** and a gearbox **564**. This winch system is mounted in a horizontal orientation such that the rotational axis is parallel with the transverse axis as opposed to the winch system shown in FIG. **12** which is oriented parallel to the vertical axis. A Person of Ordinary Skill in the Art will recognize the advantage afforded by this horizontal orientation which allows for the position and angle of the winch rope **553** coming off of the winch drum **565** to be accommodated internal to the platform portion **504** of machine **500**. Force measurement may be accomplished through the use of a force measurement device such as a load cell similar to what is shown in FIG. **12** or through the use of torque sensing instrumentation associated with servo motor **532**. While servo motors such as **532** are typically provided with position measurement devices such as encoders which allow them to keep track of their position, the output pulleys **516a**, **516b**, and **517** may be provided with encoders or equivalent hardware which allow the machine to determine which output pulley(s) are being used currently to calculate the appropriate mechanical advantage and thereby measure the correct force applied by the user. Position measurement devices may take multiple different forms including draw wire sensors, potentiometers, encoders, ultrasonic distance sensors, laser distance sensors, and many other devices known in instrumentation. As shown in FIG. **18**, winch rope **553** extends from drum **565** to a first pulley **545** passing through rollers **530** and **531** which are urged together by elastic means thereby softly grasping rope **553**. Rollers **530** and **531** may be provided with sleeve type bearing surfaces which provide a small amount of drag

which provides a small amount of tension on rope **553** as it is being wound onto drum **565**. In this way, rollers **530** and **531** along with the associated elastic element form an auto-tensioning element which assists with proper winding in addition to other features which will be discussed in greater detail. From the first pulley **545**, rope **553** extends around a second pulley **546a**, a horizontal runner pulley **547** which is on a horizontal runner block **548**. From there it passes around fourth pulley **546b** to a 2:1 splitter coupling **533**. Horizontal runner **548** has a first output rope segment **554a** which travels to the center pull location **517**. 2:1 splitter coupling **533** has a second output rope segment **554b** and a third output rope segment **554c** which pass around pulleys **549** and **550** respectively to outer pull locations **516a** and **516b** respectively. A Person of Ordinary Skill in the Art will recognize the advantages afforded by the arrangement shown in FIG. **18**. For example, machine **500** may be used for many different exercises including bench press and squats. It is well known that squats or deadlifts are the exercises that allow a user to generate their maximum force potential. On machine **500**, squats and deadlifts would be performed using center pull location **517**. In contrast to this, an exercise with lower force generating potential such as bench press will be carried out using the outer pull locations **516a** and **516b** as shown in FIG. **17**. The arrangement of ropes segments is matched to the strength characteristics of a human in terms of force capacity and speeds. Whereas the servo motor **532** need only be capable of generating sufficient force for the bench press exercise, the horizontal runner has a 2:1 mechanical advantage due to the rope **553** passing over the pulley **547**. Thus, a smaller servo motor can be used along with smaller rope in the machine which require smaller pulleys and less steel structure. Only a small segment of high-tension rope is required between the horizontal runner **548** and the center output pulley **517**. Similar to the single pulley machine shown in FIG. **11**, platform portion **504** of machine **500** may be divided into segments **560**, **561**, and **562** to permit easier transportation. Segment **560** is shown hinged to center segment **561** through hinge **563b** while segment **562** is shown hinged to center segment **561** through hinge **563a**. Wing segments **560** and **562** may be locked in orientation relative to center segment **561** through the use of locking pins **539**. The third embodiment of the present invention may be configured for transportation by removing the posts **511** (if provided), removing the locking pins **539** and folding up the wing segments **560** and **562**.

[0069] With particular reference to FIG. **19**, a detailed view is shown providing a clearer view of 2:1 splitter coupling **533** as well as horizontal runner **548**. As shown in FIG. **19**, 2:1 splitter coupling **533** comprises a set of ear plates **534** and pins **535**, **536** which connect rope **553** to ropes **554b** and **554c**. A Person of Ordinary Skill in the Art will recognize that 2:1 splitter coupling **533** allows a summation of forces on ropes **554b** and **554c** to be transmitted to rope **553** with minimal shift in longitudinal position. For instance, if ropes **554b** and **554c** have equal amounts of tension, then the 2:1 splitter coupling **533** will orient itself to an angle of 0° as shown in FIG. **19**. On the other hand, if for instance, rope **554b** has tension while rope **554c** has no tension, then 2:1 splitter coupling **53** will orient itself until the pin **536a** (connected with rope **554b**) is in-line with pin **535**. A Person of Ordinary Skill in the Art will recognize that at intermediate tension distribution conditions, 2:1 splitter coupling **533** will orient itself in intermediate angles to maintain equilibrium. As shown in FIG. **19**, horizontal runner **548** comprises a pulley **547**. Horizontal runner **548** may also be provided with wheels **538** which allow for smooth traveling in the longitudinal direction of the machine **501** when the center pull location **517** is be utilized.

[0070] With particular reference to FIG. **20**, a perspective view is provided of machine **500** showing more particulars of how the post elements **511a** and **511b** are arranged. Post elements **511a** and **511b** are mirror images of each other, and each has every feature and function as the other. However, for the sake of clarity, some features are shown on one tower and others on the other tower. Similar to the apparatus shown in FIG. **14**, rope extending from first pull location **516a** extends upward toward a vertical runner **538** which has a pulley **540**. A post rope segment **542** follows a path starting from anchor point **541** on collar **526**. From the anchor point **541** the first post rope segment **542** travels vertically to a first tower pulley **543**. From there, a second post rope

segment **544** (integral with first post rope segment **542**) travels vertically downward toward pulley **540** in the vertical runner **538**. Second post rope segment **544** wraps around vertical runner pulley **540** and then runs vertically (as post rope segment **551**) up to a second post pulley **552**. The post rope next wraps around second post pulley **552** and runs vertically downward (as post rope segment **566**) toward a third post pulley **567**. The post rope next wraps around third post pulley **567** and travels vertically upward (as post rope segment **568**) towards post swivel coupling **569**. Post swivel coupling **569** is pinned to collar **526** through the use of a structural pin **555** which allows the post swivel coupling **569** to pivot about a vertical axis centered in the structural pin **555**. Post swivel coupling is equipped with two pulleys **556**, and **557** which act to redirect post rope segment **558** depending on whether it is pulled vertically upward or downward relative to the swivel coupling **569**. A person of ordinary skill in the art will recognize that the post rope arrangement creates a 1:2 mechanical advantage over the pull force applied by the machine **500** at the first pull location **516a**. Again, this tends to match the natural capacity of a human as in doing exercises from a post structure such as **511a** and **511b** (such as is well known in the exercise field) a human naturally has lower force generating capacity but higher speed capacity relative to for instance a squat type exercise. In this way, the machine capacity at the output segment **588** will be half the capacity of the base platform but twice the speed. Each post collar **526** may be outfitted with a support cup **519** or a post swivel coupling **569**, depending on the exercise to be performed. With one or two post swivel couplings **569**, a person of ordinary skill in the art will recognize great variety in the number of different exercises that can be accommodated. To accommodate an even greater number of exercise possibilities, the machine **500** may be provided with a removable structure designed to facilitate a lat-pulldown type exercise. As shown in FIG. **20**, a lower beam **559** may be removably attached to the platform portion **504** of the machine **500**. A lap bar **571** may be adjustably attached to lower beam **559** which extends over to a centered position relative to post **511a**. Lap bar **571** may be provided with a spring-pin **572** which allows the height of the lap bar **571** to be adjusted vertically by corresponding with a set of vertically spaced holes (not shown) in the lower beam **559**. Lap bar **571**, if provided would preferentially have some means of pressure distribution to facilitate the comfort of the user during the lat pull down exercise. As shown, cushion elements **573** (consisting of a 1.sup.st and 2.sup.nd pad type supports) may be incorporated to allow the restraining force to be comfortably born by the user. Depending on the load capacity of the machine, the structure of the collars **526** and any attachments (swivel coupling **569** or support cup **519**) may have a self-weight that makes vertical adjustment of the position of collar **526** inconvenient. To the extent that this is the case, a counterweight system may optionally be provided. As shown in FIG. **20**, a counterweight system is provided beginning with a counterweight rope segment **574** which is anchored to collar **526b** at anchor point **572** and travels vertically upward toward a top pulley **573** which it wraps around and travels vertically downward through a rope port **574** which attaches to a counterweight (not shown) internal to the post tube **511b**. Optionally, machine **500** may be provided with a transport system which may have at least one wheel, shown in FIG. **20** as a set of kick-lift casters **575** or other means for facilitating the movement of the machine **500** from one location to another.

[0071] With particular reference to FIG. **21**, one embodiment of a transport system as a removable kick lift caster **575** is shown. As shown in FIG. **21**, kick lift caster **575** is shown with a frame portion **577** removably attached to the platform portion **504** of the machine **500**. The attachment may comprise a T-slot on the platform frame and a T on the transport system such that when the T is inserted into the T-slot, it can be moved upward into the stem of the T-slot to resist horizontal forces imposed when the wheel is lowered. Vertical forces are supported by the transport system bearing against the frame of the machine. A handle portion **576** is pivotally connected to the frame portion **577** via a pin **578** and is pivotable between a downward position (shown) and an upward position (not shown). Handle portion may be provided with a roller **579** which moves with handle portion **576** on an arcuate path around pin **578**. A caster **580** of ordinary design may be provided

and attached to a bar **581** which pivots on frame portion **577** about the axis of a pin **582**. A Person of Ordinary Skill in the Art will recognize that this arrangement creates a linkage mechanism which if properly dimensioned allows the roller element **579** to move past a toggle point when the handle **576** is in lowered (shown) position. Once past the toggle point (as shown) the force balance on the linkage will tend to self-support, thereby maintaining the machine **500** in an elevated position. The platform may also be selectively lowered to the ground by applying an upward force on the handle portion **576** so as to move the linkage out of the toggled condition. As shown in FIG. **20**, a preferred embodiment of the present invention may include 4 of such kick lift casters which when all put past their respective toggle positions, will completely suspend the machine **500** on the casters **580** allowing the machine to be relocated with relatively minimal effort. A person of ordinary skill in the art will recognize that many alternatives are possible to facilitate the movement of the machine **500** without departing from the scope of this disclosure. Kick-loft casters **575** may be removably attached to the machine **500** so as to keep the platform area free and clear of obstruction when the machine **500** is being transported.

[0072] With particular reference to FIG. **22**, a detailed view of the winch and servo block **583** is shown. Winch and servo block **583** comprises a servo motor **532**, a gearbox **564**, and a winch drum **565**. To aid in rope retention and to control rope winding behavior, winch drum **565** may be provided with deep flared grooved **584** which more completely guide the winch rope (not shown) **553** as it winds onto and off of the winch drum **565** into the channel formed by the groove **584**. To further aid in rope retention, the winch drum **565** may be surrounded by rope retaining covers **585** which have very narrow clearance over the winch drum **565**, thereby allowing the rope to be guided into and out of the channel formed between the cover **585** and the drum **565**. In addition to the auto-tensioner arrangement discussed previously, a person of ordinary skill in the art will recognize that the grooves **584** and the covers **585** provide additional features that aid in the proper winding of the rope **553** onto the winch drum **565**. Winch and servo block may be mounted in the frame of control tower portion (not shown) **505** through a set of hanging bars **586** and a pivot pin **587**. A retaining mechanism (not shown) may be attached to a retaining bar **588** and on its alternate end to the machine **500** to fully secure winch and servo block **583**. The retaining mechanism (not shown) maybe a load cell or a tension link.

[0073] A person of ordinary skill in the art will recognize that the present invention may be arranged in many different configurations without departing from the scope of this disclosure. For example, while the preferred embodiment may be best suited for a commercial institution such as a gym or collegiate training facility, a smaller, light-weight, and highly portable version of the present invention may be constructed as well. Such a machine may be provided with a lower maximum force capacity and be capable of being wheeled around by a single individual to be loaded into a car or truck or onto a small trailer. The lower maximum force capacity may be better suited to the common consumer and would allow the machine to be constructed to a lighter overall weight, further enhancing its mobility.

[0074] Thus, while there has been shown and described, fundamental novel features of the disclosure as applied to various specific embodiments thereof, it will be understood that 14 various omissions and substitutions and changes in the form and details of the apparatus illustrated, and in their operation, may be made by those skilled in the art without departing from the spirit of the disclosure. For example, it is expressly intended that all combinations of those elements which perform substantially the same function, in substantially the same way, to achieve the same results, are within the scope of the invention. Moreover, it should be recognized that structures and/or elements shown and/or described in connection with any disclosed form or embodiment of the invention may be incorporated in any other disclosed or described or suggested form or embodiment as a general matter of design choice. It is the intention, therefore to be limited only as indicated by the scope of the claims appended hereto.

Claims

1. A rope arrangement comprising: a thimble; a flexible tension supporting member which is terminated on the thimble; at least one protuberance, wherein the protuberance is integral with the thimble; a channel through which the tension supporting member may travel within the channel; and a stop configured to surround the tension supporting member, such that when the stop contacts the at least one protuberance of the thimble, the stop acts as a barrier to prevent the tension supporting member from traveling beyond the stop.
